

BHAVANA REDDY MALLAVARAPU

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Education

Arizona State University

Aug. 2023- May 2025

Master of Computer Science - GPA: 4.0/4.0

SASTRA Deemed University

Jun. 2016- May 2020

Bachelor of Technology in Electronics and Communication Engineering - GPA: 8.649/10.0

Technical Skills

Programming Languages: Python, Java, C, C++, JavaScript, TypeScript, Bash, HTML, SQL

Frameworks & Libraries: Node.js, Flask, Spring Boot, PyTest

Cloud & DevOps: AWS (EC2, S3, Lambda, IAM, CloudWatch, SQS, SNS), Terraform, Jenkins, Docker, Kubernetes

Monitoring & Logging: Grafana, Prometheus, Kibana

Databases: PostgreSQL, MySQL

Networking & Protocols: TCP/IP, OSPF, BGP, gNMI, NETCONF, MD-CLI, Model Context Protocol (MCP)

Version Control & Tools: Git, Postman, Linux/Unix

Relevant Courses: Distributed Systems, Operating Systems, Software Security, Data Structures & Algorithms, Statistical Machine Learning, Building RAG Agents with LLMs(NVIDIA DLI)

Experience

NOKIA

May 2024 - May 2025

Software Engineering Intern

Sunnyvale, US

- Developed asynchronous Python automation using **asyncio** for scalable orchestration of networking devices (**gNMI**, **NETCONF**, **MD-CLI**), improving system performance by **15%** and reducing testbed execution time.
- Built an end-to-end observability pipeline with **Grafana**, enhancing crash diagnostics by **50%** through real-time signal monitoring across distributed systems.
- Automated device provisioning and performance telemetry for **7220 IXR-DL** routers, increasing deployment readiness and defect detection efficiency by **20%**.
- Enhanced system resilience by refactoring application restart logic (cold vs. warm), implementing a **REST API** for real-time app tracking and boosting test integration fidelity by **30%**.
- Automated validation of **fstrim** service during OS setup, streamlining system prep by **30%** and ensuring optimal storage performance.
- Designed **CI/CD**-aware test automation infrastructure to support diverse release pipelines, increasing test coverage and bug discovery by **300%**.

Infosys Pvt Ltd

Feb. 2021 - May 2023

Senior Systems Engineer

Bengaluru, India

- Delivered high-reliability distributed services using **Kubernetes**, **Prometheus**, **Grafana**, and **Kibana** to monitor and maintain system health, reducing failure rates by **25%**.
- Automated infrastructure testing using **Terraform**, decreasing test cycles by **50%** and accelerating deployment confidence.
- Built cloud-native Lambda functions (**Node.js**) integrating **AWS S3**, **SNS**, **SQS**, **IAM**, **CloudWatch**, securing user authentication and streamlining data workflows.
- Developed performance-tracking microservices with CI/CD pipelines via **Jenkins**, improving release quality by **30%**.
- Led onboarding for junior engineers, increasing team ramp-up speed and knowledge retention by **60%**.

Awards / Achievements

Arizona State University

- Awarded the curriculum excellence for maintaining 4.0 CGPA.

Infosys Pvt Ltd

Senior Systems Engineer

- Led a team of 8 people to develop a portal to manage employee records using Java, Spring Boot, Angular and SQL.
- Recognized as one of the Top 50 Trainees (2021) for outstanding performance in the Foundation Program.
- Received the "Insta Award" for exceptional speed in acquiring the necessary technical, debugging, and analytics expertise.

Projects

Building RAG Agents with LLMs (NVIDIA DLI)

- Developed a retrieval-augmented generation (RAG) chatbot that ingests domain-specific documents, generates semantic embeddings with SentenceTransformers, stores them in Qdrant vector DB, and retrieves relevant chunks to ground GPT-4 responses, reducing hallucination by 40% and improving factual accuracy.
- Implemented an end-to-end pipeline covering document preprocessing, embedding generation, context retrieval, and prompt optimization, with a Streamlit UI for real-time interactive queries.
- Optimized retrieval latency by 35% through hybrid search (BM25 + vector similarity) and chunk size tuning, enabling sub-second response generation for datasets over 50k documents.

Distributed Systems (MIT 6.824; Instructor - Prof. Robert Morris)

Winter '24

- Implemented MapReduce, Raft, Distributed Key-Value Store.

Data Processing at Scale (CSE 511; Instructor - Prof. Samira Ghayekhloo)

Fall '24

- Analyzed Yelp's Arizona restaurant data using Hadoop and Apache Spark to derive insights on business performance and user behavior.

Engineering Blockchain Applications (CSE 598; Instructor - Prof. Swathi Punathumkandi)

Fall '24

- Developed and deployed blockchain applications using Ethereum (ERC-721 NFTs) and Hyperledger Fabric to enhance digital asset verification and supply chain transparency through smart contracts.

Cloud Computing - (CSE 546; Instructor - Prof. Yuli Deng)

Spring '24

- Developed an auto-scaling, multi-tier face recognition application on AWS EC2 with dynamic workload handling using Flask, SQS queues, multi threading and deep learning inference on App Tier.
- Built a serverless, event-driven video analysis pipeline using AWS Lambda for video frame extraction and face recognition, integrated with S3 for scalable storage.

DataBase Management - (CSE 412; Instructor - Prof. Jia Zou)

Fall '23

- Created a demo application for a patient to book appointment with doctor and for doctor to view appointments and patient health history with PostgreSQL.

Driver Action Recognition System - (Instructor - Prof. Narasimhan K.)

Spring '20

- Designed prototype employing CNN and ANN algorithms with python and GUI to identify action of driver of vehicle and alert by means of Arduino with LED and buzze.
- Designed prototype to identify hand gesture with MATLAB based on convex hull algorithm.